COLLEGE OF INFORMATION SCIENCE

CYBR312
Python Scripting in Cybersecurity
Week 6: Python Logging

Cameron Covington • Adjunct Instructor

Week Six Agenda

Week 6

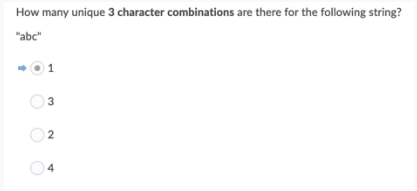
- Week 5 Review
- Introduction of Python Classes / Objects
- Leveraging Python Logging Objects
- Assignment: HashCreation and Verification Script

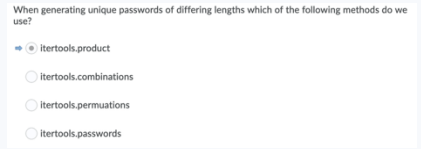


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Quiz 5 Review

Use this slide as a visual reference during the lecture.







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WK-5 Solution Demo

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Week Six Assignments

Week 6

- Your To Do List
- Attend: Week Six Live Lecture
- Quiz 6: Cyber Operations Logging
- Assignment 6: Utilize the cyber security script provided by the professor and add the ability to generate a detailed log of the script operation.

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Videos

Week 6

- Week Six Videos
- This week's video will be presented live during the Webinar.

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Key Ideas

Week 6

- Topics
- Python Logging Library
- Why is logging important in Cyber Operations?

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Python Logging Library

- Why Logging?
- When creating scripts for cybersecurity, forensics and/or incident response logging events and details is vital.
- The Built-in Python Logging Module
- Defines functions and classes which create a flexible event logging system for your scripts.

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Visual example: Week 6

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Using the Logging Library

```
'''
Python Logging Example
Professor Hosmer
August 2020
'''
import logging  # Python standard logging library
print("\nSample Python Logging\n")
logging.basicConfig(
    filename="Script.log",
    level=logging.DEBUG,
    format="%(asctime)s %(message)s"
)
logging.info("Script Started")
logging.warning("Sample Warning Message")
logging.error("Sample Error Message")
logging.critical("Sample Critical Message")
print("\nScript Complete")
```

The logging level and message are written to Script.log with a timestamp.

```
2020-08-10 10:35:08,187 Script Started
2020-08-10 10:35:08,188 Sample Warning
Message
2020-08-10 10:35:08,188 Sample Error
Message
2020-08-10 10:35:08,188 Sample Critical
Message
```

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Using the Logging Library

```
'''
Python Logging Example
Professor Hosmer
August 2020
'''
import logging      # Python Standard Logging Library
print("\nSample Python Logging\n")
logging.basicConfig(filename='Script.log',level=logging.DEBUG,form
at='%(asctime)s %(message)s')
logging.info('Script Started')
logging.warning('Sample Warning Message  ')
logging.error('Sample Error Message    ')
logging.critical('Sample Critical Message ')
print("\nScript Complete")
```

```
2020-08-10 10:35:08,187 Script Started
2020-08-10 10:35:08,188 Sample Warning Message
2020-08-10 10:35:08,188 Sample Error Message
2020-08-10 10:35:08,188 Sample Critical Message
```

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Using the Logging Library

```
'''
Python Logging Example
Professor Hosmer
August 2020
'''
import logging      # Python Standard Logging Library
print("\nSample Python Logging\n")
# Turn on Logging
logging.basicConfig(filename='ScriptLog.txt',
level=logging.DEBUG,
format='%(asctime)s - %(process)d- %(levelname)s- %(message)s')
# Sample usage
logging.info('Script Started')
logging.warning('Sample Warning Message  ')
logging.error('Sample Error Message    ')
logging.critical('Sample Critical Message ')
print("\nScript Complete")
```

Sample Output

```
1
2 2020-10-12 10:04:58,398 - 14100- INFO- Script Started
3 2020-10-12 10:04:58,398 - 14100- WARNING- Sample Warning Message
4 2020-10-12 10:04:58,399 - 14100- ERROR- Sample Error Message
5 2020-10-12 10:04:58,399 - 14100- CRITICAL- Sample Critical Message
```

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Questions?

Week 6

Coming Next Week

Introduction to Image Metadata Extraction and JSON and CSV Libraries



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